

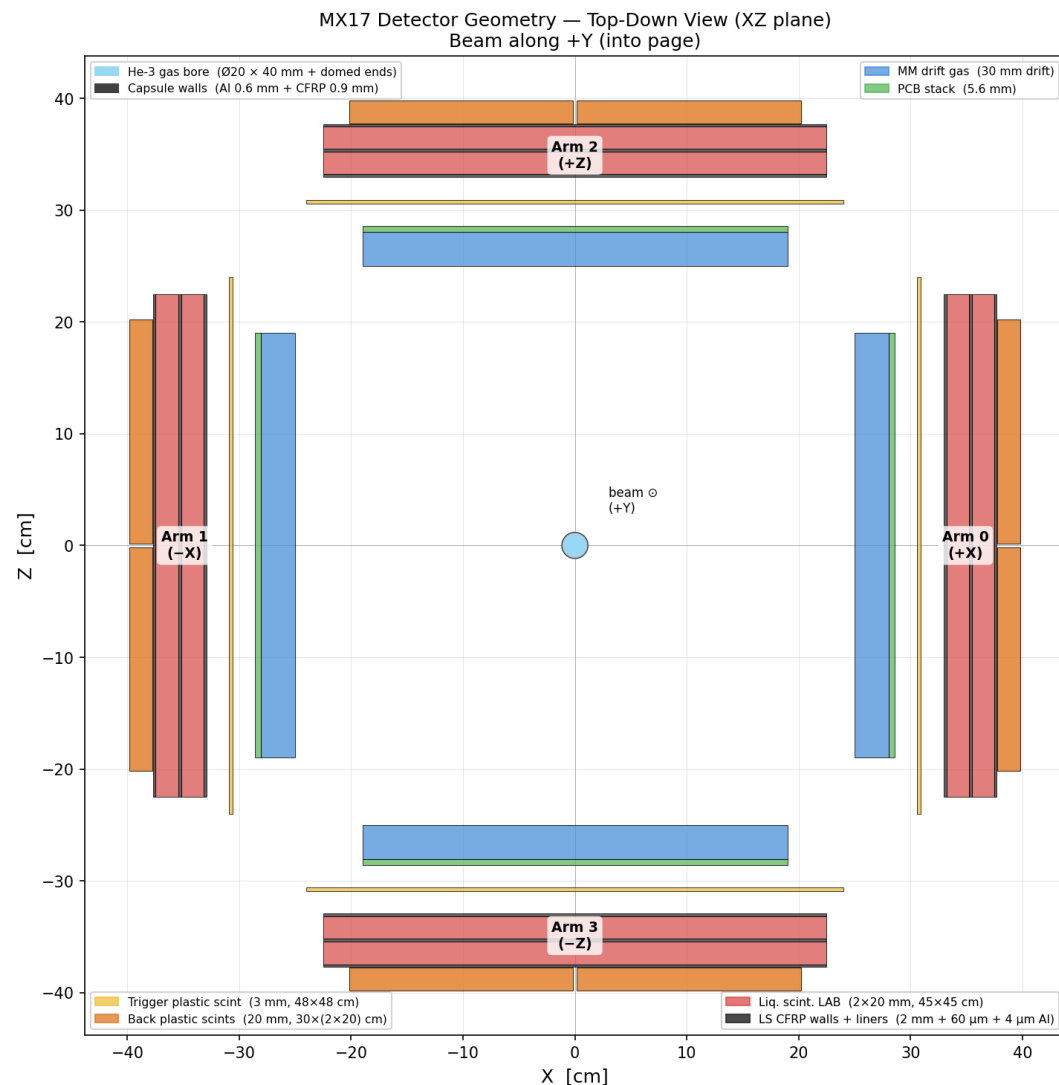
The X17 search at n_TOF EAR2

Full Geant4 simulation: production, acceptance, and the path to a measurement
Final July setup · expected results · the post-LS3 upgrade

D. Neff — 16 June 2026 | simulation + analysis: this repository (MX17_Full_Geant)

The experiment: a 4-arm $e^+ e^-$ pair spectrometer

${}^3\text{He}(n,\gamma){}^4\text{He}^* \rightarrow (\text{X17 or IPC}) \rightarrow e^+ e^-$; measure the opening angle in 4 Micromegas + scintillator arms



Target

${}^3\text{He}$ gas capsule, 500 bar, $r = 10$ mm, 60 mm on axis; thin Al (0.6 mm) + CFRP (0.9 mm) wall.

Tracking

4 Micromegas arms, front face ≈ 22 cm from the target, 30 mm drift gap — the $e^+ e^-$ directions.

Calorimetry / trigger

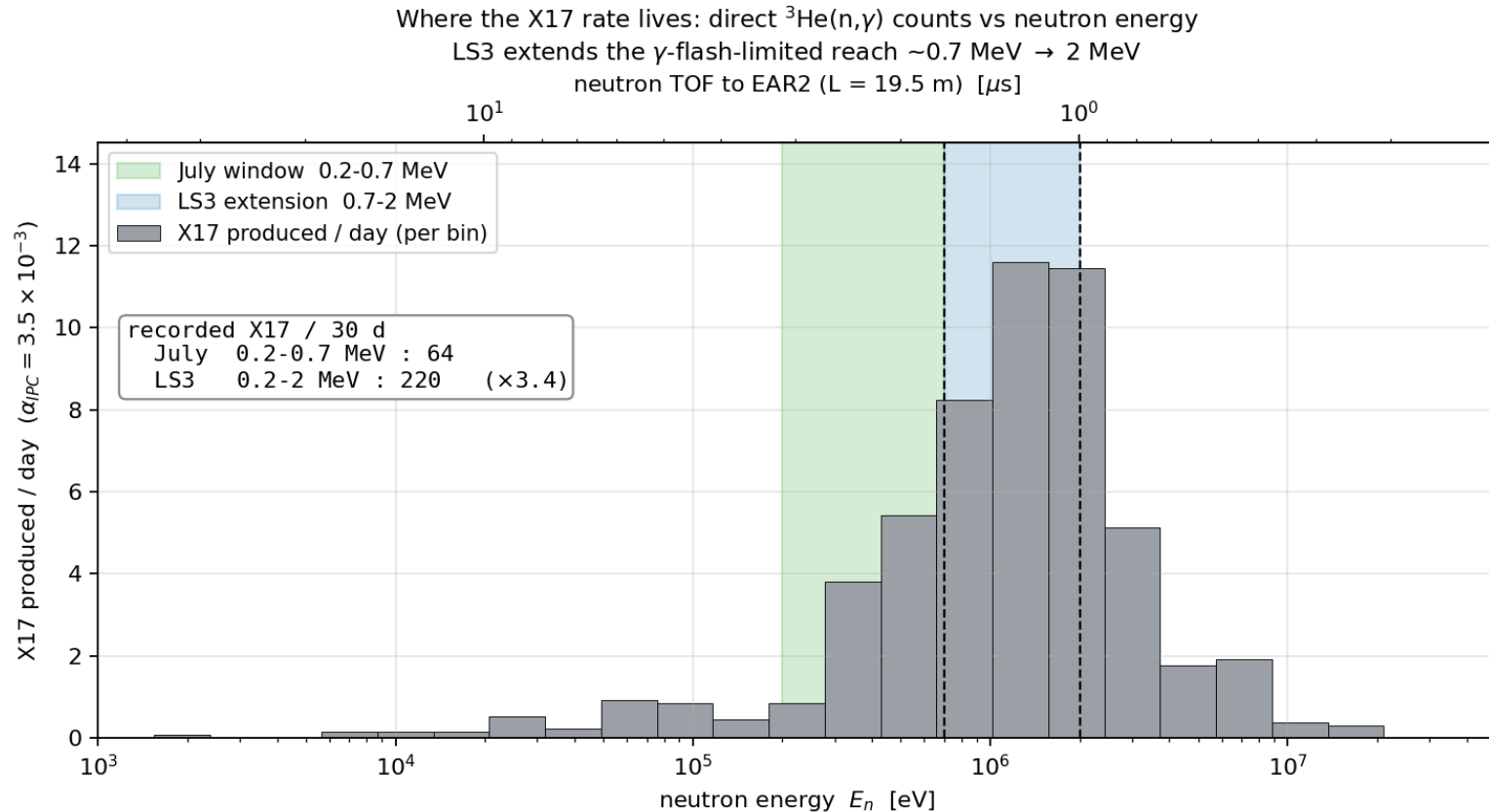
Liquid-scintillator + plastic layers per arm (assumed saturated/unusable for the high-energy measurement — tracking-only baseline).

The observable

X17 ($m \approx 17$ MeV) makes a sharp opening-angle shoulder at $\theta \approx 109^\circ$; the IPC continuum is the irreducible background.

Production is settled — and it lives at MeV energies

Directly counted ${}^3\text{He}(n,\gamma)$ from 5×10^8 neutrons; the γ -flash sets the high-energy reach



Sub-keV is dead.

The gas is opaque to (n,p); radiative capture is capped at $\sigma_n \gamma / \sigma_{np} \approx 10^{-8} \rightarrow \sim 0.1$ X17/day.

The MeV window is the measurement.

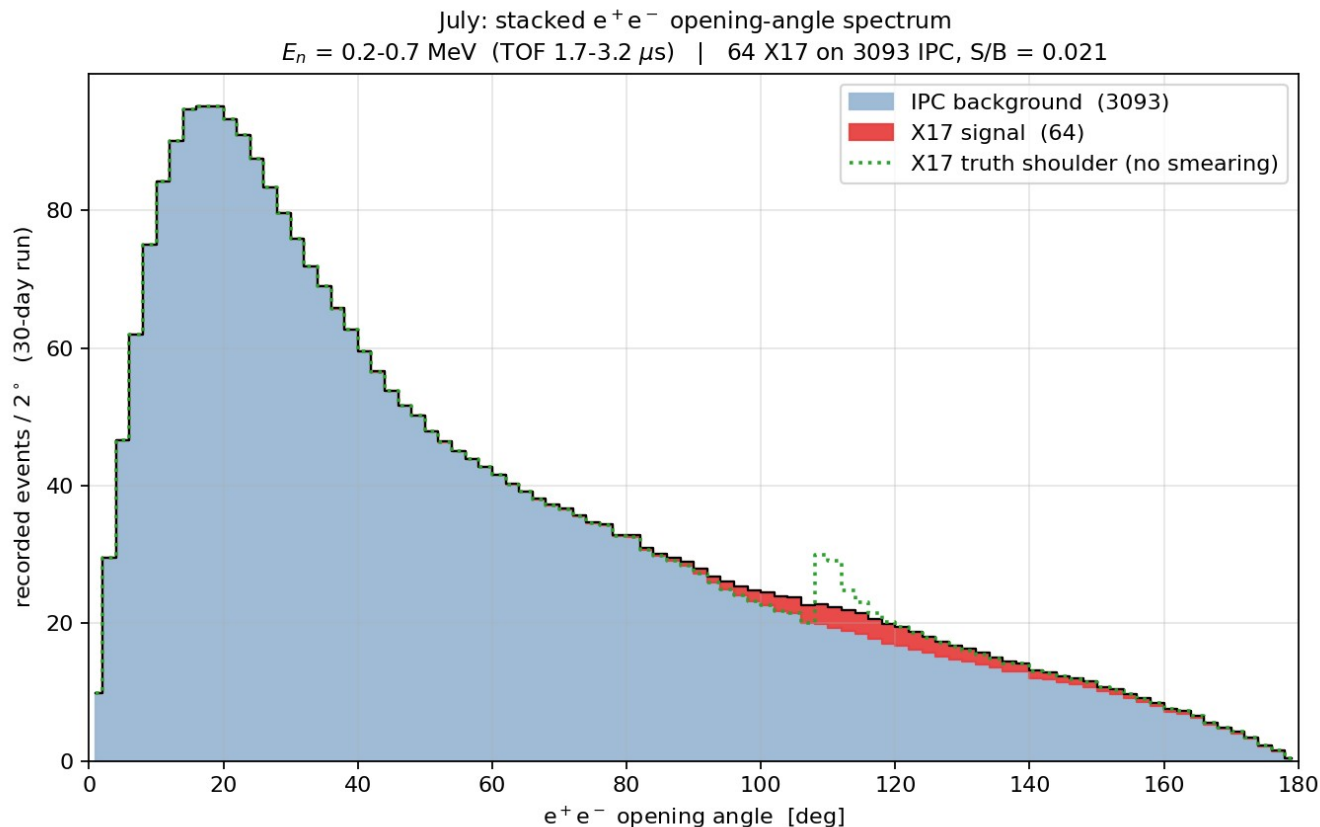
Above ~ 100 keV the gas is thin and the rate table is confirmed by Geant4 (within 10-70%). ~ 33 X17/day produced over the full range.

The γ -flash limits the reach.

Faster neutrons arrive sooner, closer to the flash. How early the detectors recover sets the max neutron energy — this is the LS3 lever.

July run: expected result (optimistic scenario)

Window 0.2–0.7 MeV — conservative γ -flash recovery (detectors usable by $\approx 1.7 \mu\text{s}$); 30-day run



$\approx 64 \text{ X17}$ on $\approx 3 \text{ 100 IPC}$

recorded in 30 days (S/B ≈ 0.021).

The peak is diluted, not visible by eye.

Capsule multiple scattering smears θ by $\sigma_{68} \approx 14^\circ$; the 109° shoulder (green dotted = truth) becomes a broad bump. The measurement is a template fit, not a cut-and-count.

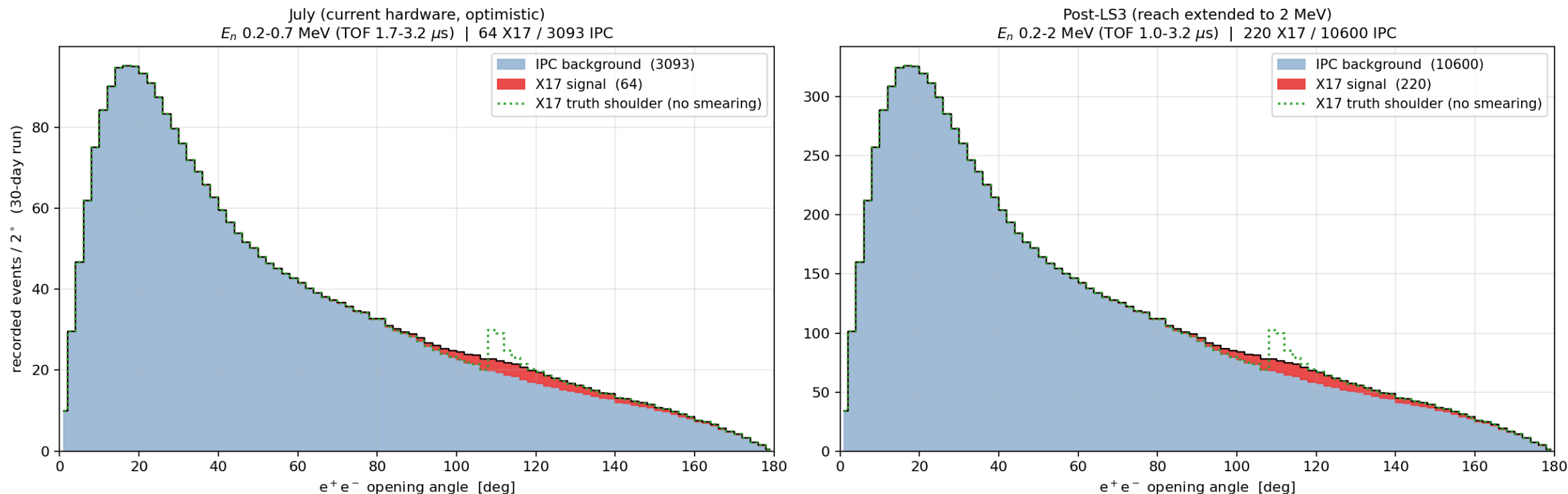
Caveats / assumptions (separate box — delete before INFN if desired)

- Optimistic: best-estimator (target-centre chord) smearing; no pile-up, no γ -flash losses, no θ -dependent MM acceptance shaping.
- $\alpha_{\text{IPC}} = 3.5 \times 10^{-3}$ ($^{12}\text{C}/^8\text{Be}$ -anchored) and X17/IPC = 2.5% are assumed inputs (provenance pending). Table $\alpha = 2.1 \times 10^{-3}$ would scale signal $\times 0.6$ ($\rightarrow \sim 38 \text{ X17}$).
- At-rest kinematics (shoulder shifts a few degrees with the E_n boost). MM-double acceptance 19.6% (X17) / 23.6% (IPC).

After LS3: extend the reach to 2 MeV neutrons

Faster γ -flash recovery ($\approx 1.0 \mu\text{s}$) $\rightarrow$ window 0.2–2 MeV; same analysis, $\times 3.4$ the statistics

Expected recorded opening-angle spectrum: July vs post-LS3 ($\alpha_{IPC} = 3.5 \times 10^{-3}$, MM-double acceptance, best-estimator smearing)



July 64 X17 $\rightarrow$ post-LS3 220 X17 recorded (30 d), on the same $S/B \approx 0.021$.

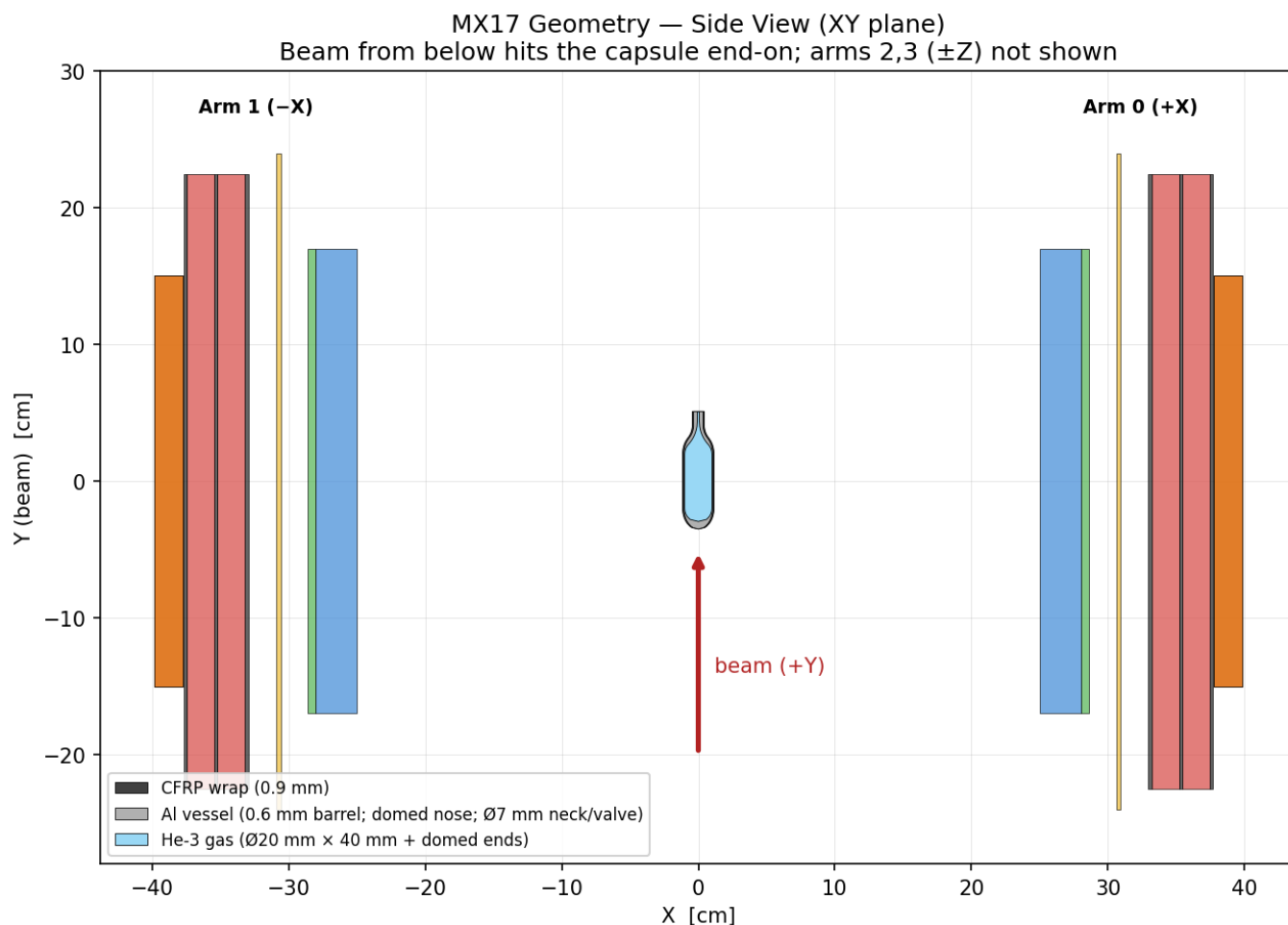
Reaching 2 MeV neutrons (faster γ -flash recovery) adds the most productive energy decade. The spectrum shape is unchanged — the gain is purely statistical, which is exactly what a template fit converts into significance.

Supporting results

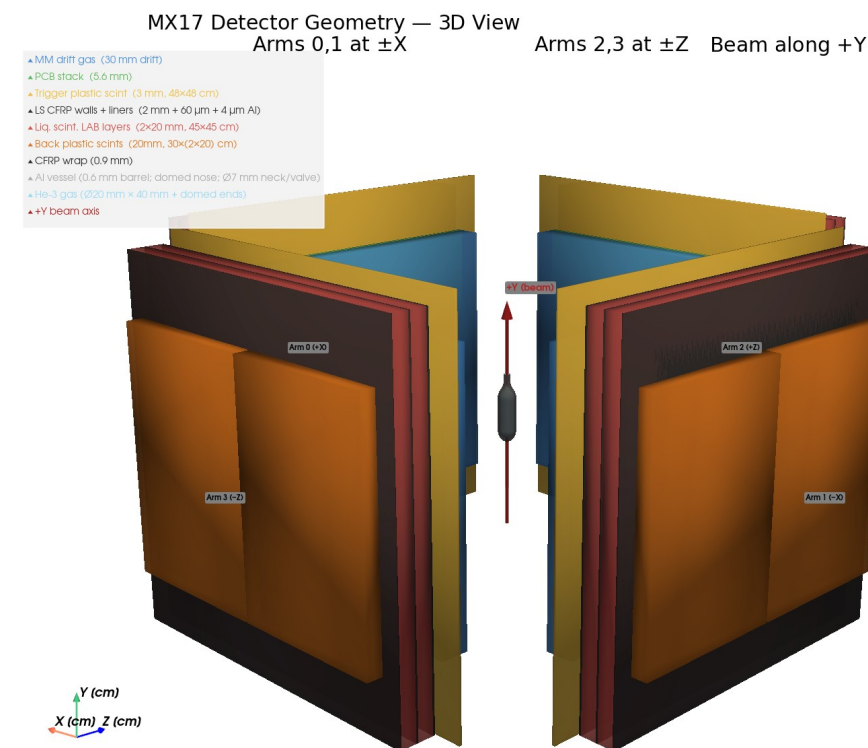
Production budget · the angular-resolution floor · the analysis strategy · the γ -dark E0 channel · backgrounds

Detector geometry — side view and 3D

Companion views to the top-down layout: the ^3He capsule shape, beam-end-on incidence, and the full 4-arm arrangement



Side view (XY): beam enters end-on along +Y into the capsule; arms 0/1 ($\pm X$) shown.



3D: 4 arms ($\pm X$, $\pm Z$), each MM + scintillator stack around the target.

The angular resolution is hardware-frozen

Opening-angle resolution is set by multiple scattering in the capsule wall — no reconstruction can beat it

$$\sigma_{68}(\Delta\theta) \approx 14.5^\circ$$

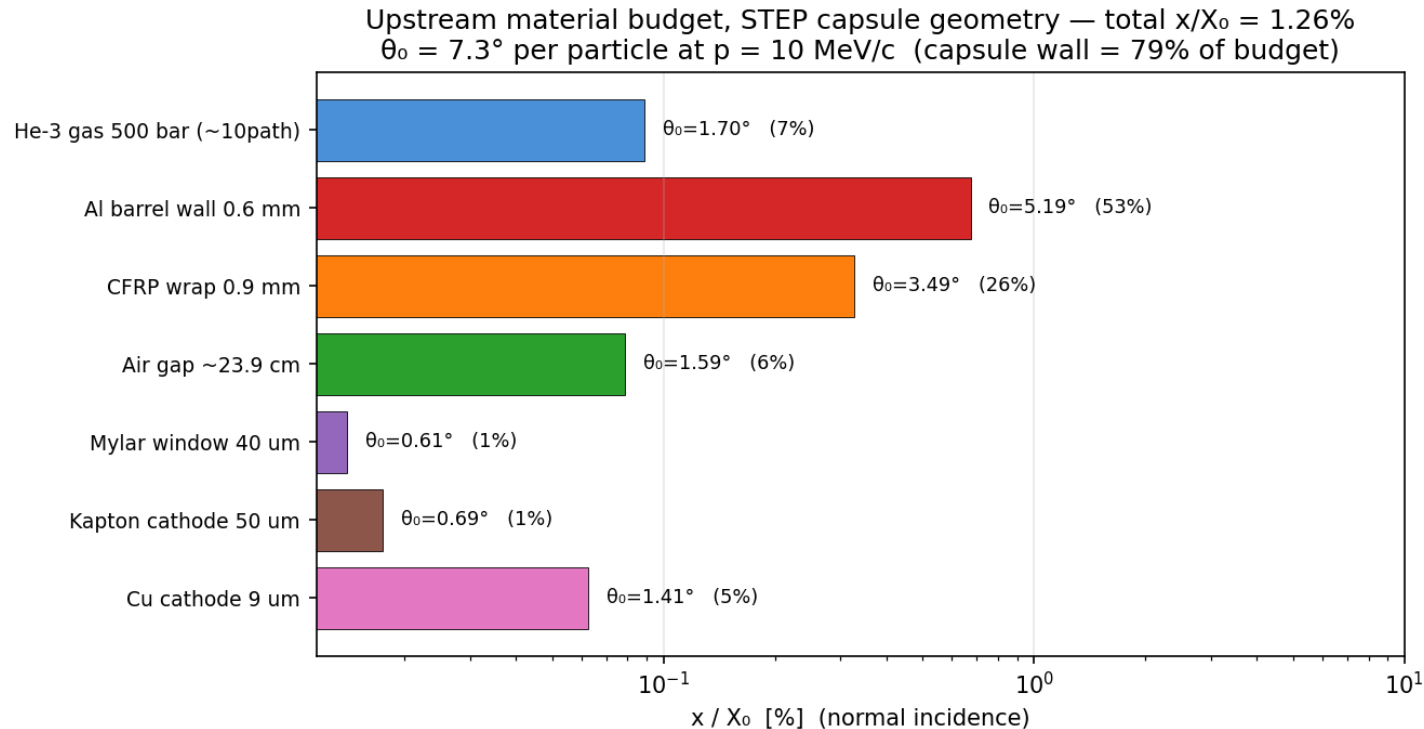
full accepted spectrum ($\approx 11.5\text{--}12.5^\circ$ for symmetric pairs), even assuming a perfect vertex.

Why:

The 0.6 mm Al + 0.9 mm CFRP wall (~ 11 mm from the vertex) is 79% of the material budget. A scatter that early rotates the whole visible track.

Verdict:

All 7 reconstruction methods land within 2° of each other. The only hardware lever is an all-composite vessel ($\rightarrow \sim 10^\circ$). The recovery path is statistical: fit the shape.

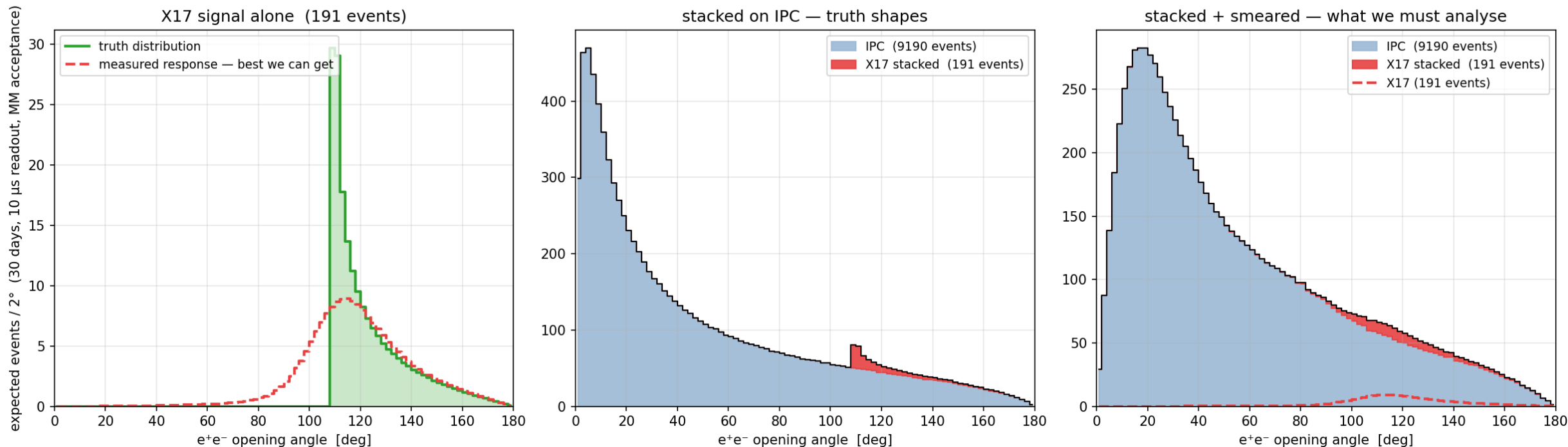


Why the analysis is a template fit, not a peak cut

Truth → stacked on IPC → both smeared: the shoulder washes out, but its shape is known to $\sim 1\%$ from 10^7 simulated pairs

Best we can do with capsule multiple scattering — 30-day recorded yields (10 μ s readout, MM-double acceptance)

BEST CASE: per-leg smearing from the measured $P(\psi|KE)$ response, best estimator (target-centre chord); production-level shapes, no pile-up backgrounds, no γ -flash losses

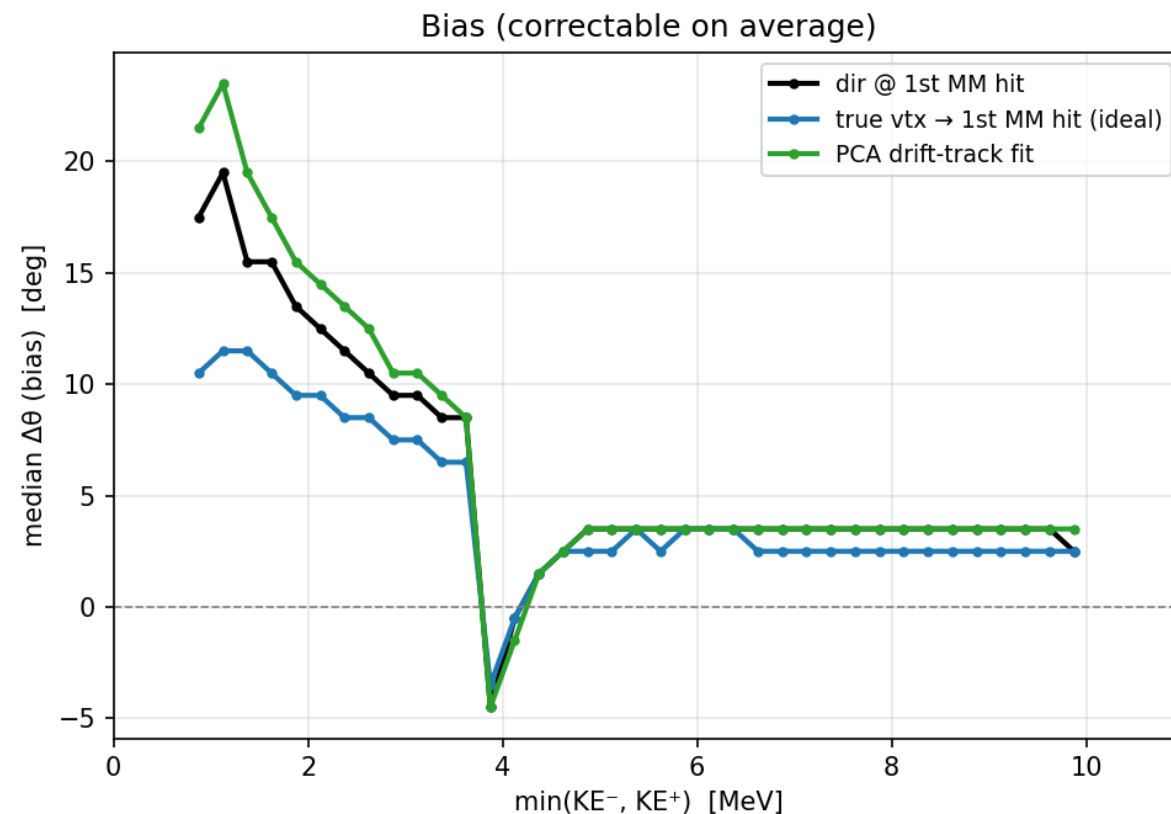
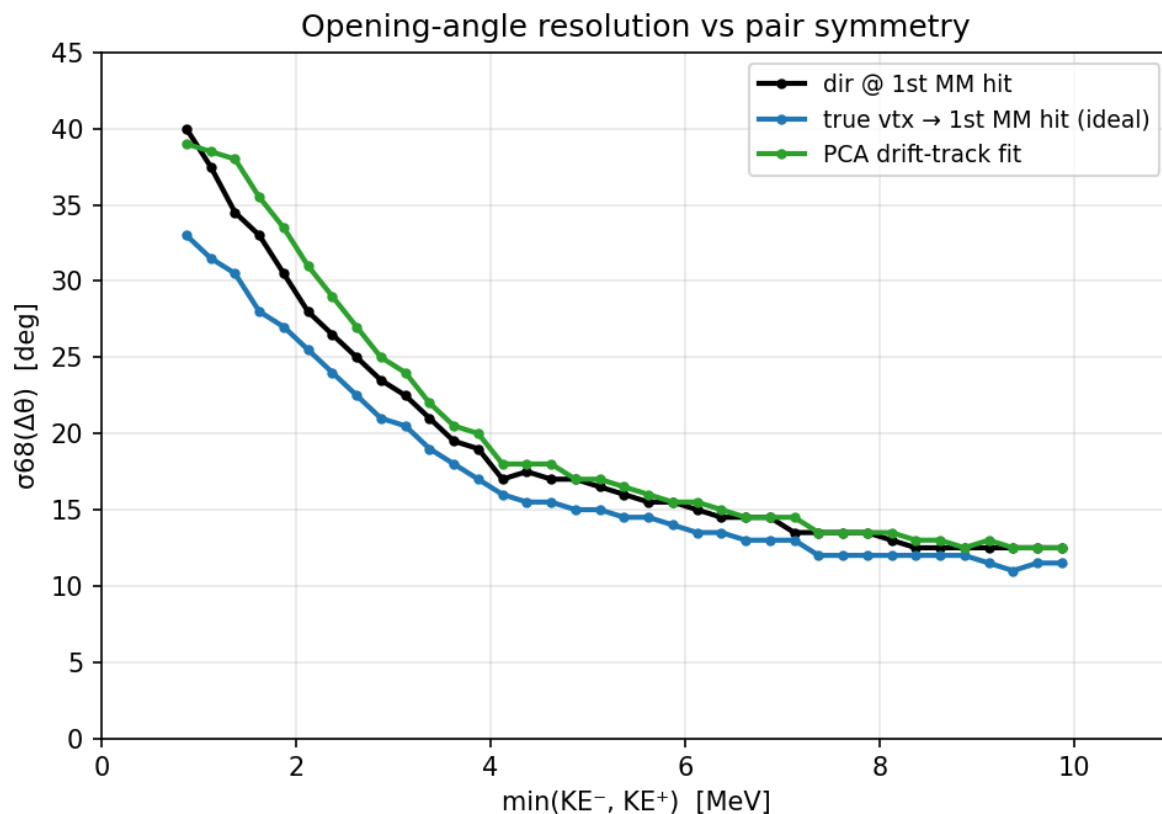


Cut-and-count on θ is dead (recorded $S/B \approx 0.02$). What survives: a fit of the known smeared-signal template + the smooth IPC continuum, with resolution entering as dilution of statistical power rather than as a cliff. This is the natural go/no-go calculation.

Resolution is a soft-track problem

σ_{68} vs pair symmetry, and the +bias (correctable in a response-matrix fit) — X17 is sharpest at its symmetric 109° shoulder

Pair opening-angle $\Delta\theta = \theta_{\text{reco}} - \theta_{\text{truth}}$ vs min pair KE (X17 + IPC combined)

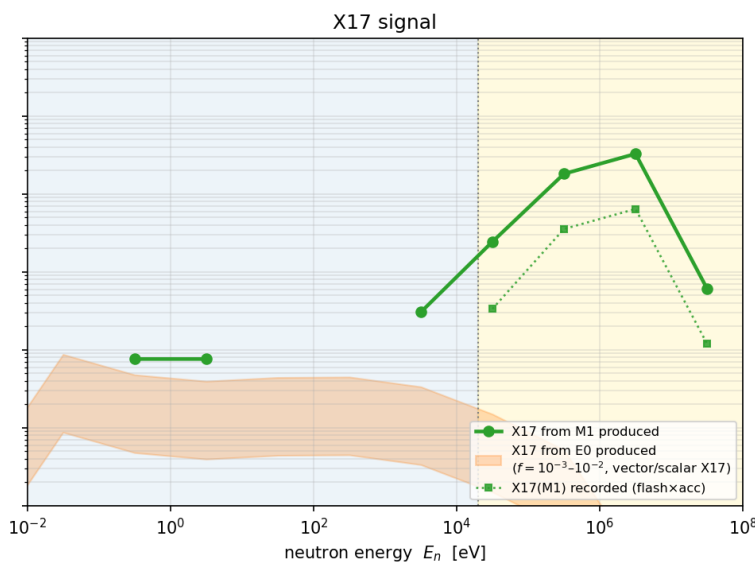
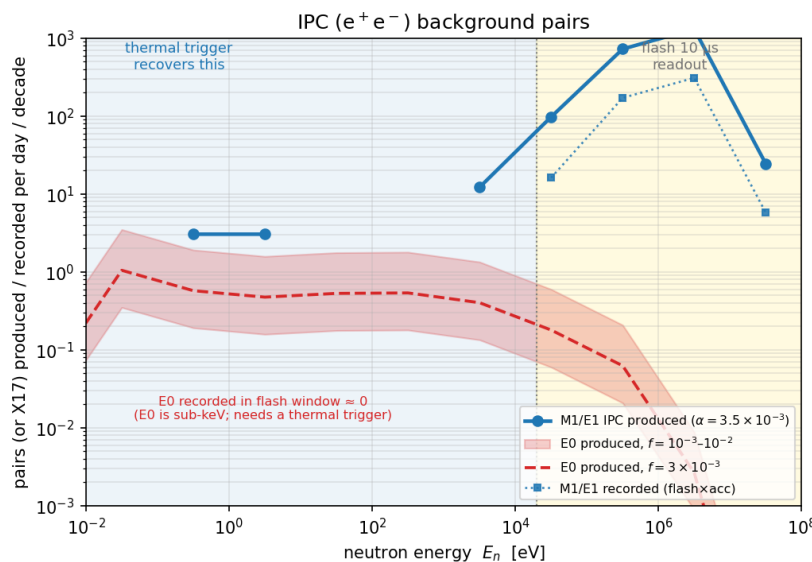


Below ~ 3 MeV a leg is directionless after the wall ($\sigma_{68} \rightarrow 40^\circ$); above ~ 8 MeV it plateaus at $\sim 12^\circ$. X17 is sharpest at its symmetric 109° shoulder — the good news.

The γ -dark E0 channel: completeness, not a game-changer

$0^+ \rightarrow 0^+$ captures make no photon but can make a massive X17 — yet the rate is tiny and sub-keV

Final metric: X17 / IPC pairs vs neutron energy, with the γ -dark E0 channel and the trigger window
flash 10 μ s readout reaches $E_n \gtrsim 20$ keV; E0 lives sub-keV $\rightarrow$ only a thermal trigger records it



E0 is real but small.

The 0^+ doorway sits sub-keV, where the same 10^{-8} radiative branching that kills the thermal program applies — times $f \leq 10^{-2}$.

The flash readout cannot see it.

Sub-keV $\rightarrow$ TOF $> 45 \mu$ s $\rightarrow$ arrives after the window closes. Only a dedicated thermal trigger records it (≤ 0.06 X17/day).

So:

Leaving E0 out does not bias the baseline high-energy analysis. Worth adding to the generator for completeness only.

Summary & status

Production is settled.

Full Geant4 (5×10^8 neutrons) confirms the rate table where it is valid: ~ 33 X17/day produced, living at MeV energies. The sub-keV measurement is dead (self-shielding).

July (current hardware): ~ 64 recorded X17 / 30 d.

Window 0.2–0.7 MeV under a conservative γ -flash recovery; on ~ 3 100 IPC ($S/B \approx 0.02$). Extractable by a template fit, not a peak cut.

Post-LS3: ~ 220 recorded X17 / 30 d.

Reaching 2 MeV neutrons ($\times 3.4$ statistics). The decisive measurement is the real γ -flash recovery time.

The resolution is hardware-frozen at $\sigma_{68} \approx 14^\circ$.

Capsule wall MS; an all-composite vessel is the only lever. E0 is a completeness item, not a driver.

Open items:

α_{IPC} & X17/IPC provenance (Alberto); E_n -dependent generator kinematics; the template-fit significance projection; γ -flash recovery measurement.